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 Course: CSE423 Section: 18

Ans to the Q No-1

(a)

Here, $(5, -15)$ to $(0, -5)$

$$\text{So, } m = \frac{-5 + 15}{0 - 5} = -2$$

$$\text{So } \frac{1}{m} = \frac{1}{-2} = -0.5$$

We know

if $-1 < m < 1$ then,

$$y_k = y_{k-1} + m \text{ and } x_k = x_{k-1} + 1$$

else,

$$x_k = x_{k-1} + \frac{1}{m} \text{ and } y_k = y_{k-1} + 1$$

y_{k+1}	$x_{k+1} (+\frac{1}{m})$	x (round)	Pixel
-15	5	5	(5, -15)
-14	4.5	5	(5, -14)
-13	4	4	(4, -13)
-12	3.5	4	(4, -12)
-11	3	3	(3, -11)
-10	2.5	3	(3, -10)
-9	2	2	(2, -9)
-8	1.5	2	(2, -8)
-7	1	1	(1, -7)
-6	0.5	1	(1, -6)
-5	0	0	(0, -5)

[Initial Point]

So the pixels are $(5, -14), (4, -13), (4, -12), (3, -11), (3, -10), (2, -9), (2, -8), (1, -7), (1, -6), (0, -5)$

Ans)

(b) (5, -15) to (0, -5).

$$dy = -5 + 15 = 10$$

$$dn = 0 - 5 = -5$$

So the value is from Zone-2.

Now

Zone-2 to Zone-0,

$u = Y$ $g = -X$
 (5, -15) will be (-15, -5)
 (0, -5) will be (-5, 0)

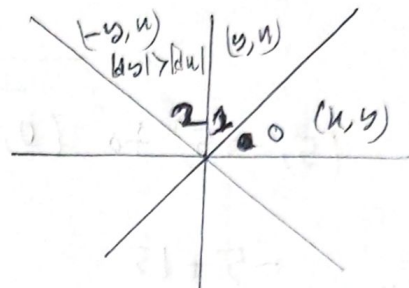
$$dy = 0 + 5 = 5$$

$$dn = -5 + 15 = 10$$

$$d_{int} = 2dy - dn = 0$$

$$\Delta d_{NE} = 2dy - 2dn = -10$$

$$\Delta d_E = 2dy = 10$$



Serial	x	y	NE or E	d	Original zone	
					$x = -y$	$y = x$
1	-15	-5	E	$0 + 10 = 10$	5	-15
2	-14	-5	NE	$10 - 10 = 0$	5	-14
3	-13	-4	E	$0 + 10 = 10$	4	-13
4	-12	-4	NE	$10 - 10 = 0$	4	-12
5	-11	-3	E	$0 + 10 = 10$	3	-11
6	-10	-3	NE	$10 - 10 = 0$	3	-10
7	-9	-2	E	$0 + 10 = 10$	2	-9
8	-8	-2	NE	$10 - 10 = 0$	2	-8
9	-7	-1	E	$0 + 10 = 10$	1	-7
10	-6	-1	NE	$10 - 10 = 0$	1	-6
11	-5	0	E	$0 + 10 = 10$	0	-5

Ans to the Q No-2

(a) The AND operation is used in Cohen Sutherland algorithm to quickly detect trivial rejection of a line. If both endpoints have a nonzero result after performing a bitwise AND on their region code, the line is completely outside and rejected.

(b) $P_0(1,3)$ and $P_1(7,10)$

Now,

$$D = (x_1 - x_0, y_1 - y_0) = (7 - 1, 10 - 3) = (6, 7)$$

We know,

$$P(t) = P_0 + t \times D$$

$$\therefore P(0.3) = (1, 3) + (0.3)(6, 7)$$

$$= (1 + 1.8, 3 + 2.1)$$

$$= (2.8, 5.1)$$

(Ans)

(c) (i) $x_{\min} = -50$ $y_{\min} = -10$

$x_{\max} = 60$ $y_{\max} = 100$

Here, $A(-50, 70)$ to $B(40, 100)$

1001	1000	1010
0001	0000	0010
0101	0100	0110

outside 1 = ~~0000~~ 0100

outside 2 = 0000

Now 0100

AND 0000

0000

So the line is partially inside. As it's 0100 so, $y = y_{\min}$.

Now $m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{100 - 70}{40 - (-50)} = \frac{17}{9}$

So $y - y_1 = m(x - x_1)$

$x = x_1 + \frac{1}{m}(y - y_1)$

$= -50 + \frac{9}{17}(-10 + 70) = (-18.23)$

So, New point $S(-18.23, -10) \approx (-18, -10)$

Recalculated outside 1 = 0000

Previous outside 2 = 0000

So now the value is completely inside the window.

It will be clipped to $(-18, -10)$ to $(40, 100)$
(Am)

(ii) Here,

$$x_{\min} = -50$$

$$y_{\min} = -10$$

$$x_{\max} = 60$$

$$y_{\max} = 100$$

Here $P_0(-50, -70)$ and $P_1(40, 100)$

So,
 $D = (x_1 - x_0, y_1 - y_0) = (40 + 50, 100 + 70) = (90, 170)$

Initially $TE = 0$ and $TL = 1$.

Boundary	N	N·D	t	PE/PL	TE	TL
Left	(-1, 0)	-90	$\frac{x_{\min} - x_0}{x_1 - x_0} = 0$	PE	✓ 0	1
Right	(1, 0)	90	$\frac{x_{\max} - x_0}{x_1 - x_0} = 1.22$	PL	0	✓ 1
Bottom	(0, -1)	-170	$\frac{y_{\min} - y_0}{y_1 - y_0} = 0.353$	PE	✓ 0.353	1
Top	(0, 1)	170	$\frac{y_{\max} - y_0}{y_1 - y_0} = 1$	PL	0.353	✓ 1

Here $t > TE$ so the line is partially inside we have to clip it.

So, For $t = 0.353$,

$$P(0.353) = (-50, -70) + (0.353) \times (90, 170) = (-18.23, -9.99) \\ = (-18, -10)$$

For $t = 1$,

$$P(1) = (-50, -70) + (1) \times (90, 170) = (40, 100)$$

So, $(-18, -10)$ and $(40, 100)$ are the end points of the clipped line. (Ans)